(6)

EE-B-002181A(Army-GL)
December 30, 1985
SUPERSEDING
EE-B-002181(Army-GL)
March 18, 1983

INTERIM FEDERAL SPECIFICATION

BEEF, ROAST, WITH MUSHROOM GRAVY, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers roast beef slices with mushroom gravy thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, For Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

AMSC N/A FSC 8940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC: Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

U.S. Standards for Grades of Canned Mushrooms

(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Dehydrated Onion and Garlic Association (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association 375 Sutter Street, Suite 700, San Francisco, CA 94108.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of recognizable primal or subprimal cuts of the round, shank-off (round, top round, bottom round, and knuckles). Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. The beef shall be in the fresh-chilled state and shall be in excellent condition, i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat which has been properly stored and handled; cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than a slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

c. Holding in the frozen state for not more than 120 days after placement in the freezer is permitted provided that the beef is:
-placed in the freezer within 4 hours after certification
-frozen to 0°F or lower within 72 hours after placement in the freezer
-stored at 0°F or lower
-protected from freezer deterioration and damage
-stored in containers that are adequate to maintain product excellence
-held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

3.2.2 Mushrooms, canned, stems and pieces. Canned mushroom stems and pieces shall be U.S. Grade B or better in accordance with the U.S. Standards for Grades of Canned Mushrooms.

3.2.3 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized modified maize food starch for use in canned foods. The modified starch shall partially gel at temperatures in the range 140° to 190°F and fully gel at common retort temperatures, and shall be resistant to low pH, shear stress, and have excellent cold storage stability. In addition, the starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900. The starch suspension shall be visually opaque after cooking and thermal processing (retorting).

3.2.4 Shortening, vegetable. The vegetable shortening shall be refined, hydrogenated, deodorized vegetable oils, having a stability of not less than 100 hours (AOM). The shortening shall have a uniform plastic texture and be free from rancid, sour, or other objectionable flavors and odors. The shortening may contain antioxidants (active ingredients BHA, BHT, TBHQ, or propyl gallate) alone or in combination and antifoaming agents or a combination of both in amounts as permitted by the Food and Drug Administration.

3.2.5 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anticaking agents.

3.2.6 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

3.2.7 Sugar, granulated. Sugar shall be refined, sucrose, granulated cane or beet or a combination of both and shall comply with MIL-STD-900.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.2.8 Vegetable protein, hydrolyzed (HVP). The hydrolyzed vegetable protein shall be obtained by the hydrolysis of any one or any combination of corn, soybean, or wheat proteins from which none of the monosodium glutamate has been extracted during processing and shall have a characteristic color and flavor. Salt (chloride calculated as NaCl) shall not exceed 35 percent. The pH of a 3-percent solution shall not be less than 5.0 nor more than 6.3. Hydrolyzed protein shall be soluble in water at 90°F to the extent of not less than 5 grams in 100 milliliters.

3.2.9 Monosodium glutamate. Monosodium glutamate (MSG) shall comply with the Food Chemicals Codex.

3.2.10 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.11 Caramel color. Caramel color shall comply with the Food Chemicals Codex.

3.2.12 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.13 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.14 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.2.15 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Roast beef preparation. The roast beef shall be prepared as follows:

a. Fifteen percent of the boned and trimmed beef shall be mechanically ground twice through a 1/8 inch grinder plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

b. The balance of the boned and trimmed beef shall be mechanically ground once through a kidney-shaped plate having openings measuring 2½ inches or more in the least dimension using a two-bladed knife that yields pieces weighing at least 2 ounces.

3.3.1.1 Mixing. The ground meat shall be mechanically vacuum mixed with the following ingredients in the proportions indicated. Time from grinding to mixing shall not exceed 4 hours. Mixing shall continue for approximately 15 minutes or until the product becomes moderately tacky.

Ingredient	Percent by weight
Beef	96.50
Water (or ice)	2.00
Salt	1.00
Sodium tripolyphosphate	0.50

3.3.1.2 Stuffing and forming. The mixed meat shall be mechanically stuffed tightly into number 6½ regenerated cellulose prestuck casings. The open ends shall be left temporarily opened. The resultant meat logs (rolls) shall be pressed between spring-loaded screens separated by 1-3/4 inch spacers, such that rectangular shaped logs are formed measuring 1-3/4 inch ± 1/8 inch in height. The open ends shall then be sealed exerting minimum pressure on the meat. The stuffed meat logs shall be in the cooking process within 4 hours after being stuffed.

3.3.1.3 Cooking. The pressed meat logs in the screens shall be cooked in a cook house (smoke house without smoke; see 6.4) or by another commercially acceptable cook method to provide a product meeting the finished product drained weight and other requirements.

3.3.1.4 Cooling and freezing. The cooked meat logs shall be cooled to 40°F or below within 4 hours following the cooking process. The cooled, cooked meat logs shall be held not longer than 48 hours and at a temperature not to exceed 40°F after removal from the cooking process and prior to slicing. Alternatively, if the cooked meat logs are to be held longer than 48 hours after cooking, they shall be placed in a freezer within 4 hours after cooking and frozen to 0°F or lower within 24 hours after being placed in the freezer. In the case of the alternate method, the meat shall not have been previously frozen and the meat shall be mechanically vacuum stuffed into heat-sealable casing materials made from FDA and USDA food approved laminate consisting of 0.0005-inch thick polyester and 0.0015-inch thick medium density polyethylene. The frozen cooked meat logs shall be held at 0°F for a period not more than 30 days prior to slicing. The frozen meat logs shall be tempered to a range of 28° to 40°F for slicing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

3.3.1.5 Slicing of meat logs. The tempered cooked meat logs shall be sliced into slices having uniform size and shape to yield approximately 8 slices per pound. The time from slicing to filling shall be not more than 4 hours.

3.3.2 Gravy preparation. The gravy shall be prepared from the following ingredients:

Ingredient	Percent by weight
Water	73.38
Mushrooms, canned, stems and pieces	14.00
Starch, modified, high opacity	5.50
Shortening, vegetable	4.50
Sugar, granulated	0.50
Vegetable protein, hydrolyzed	0.50
Monosodium glutamate	0.50
Salt	0.50
Lecithin	0.32
Caramel color	0.10
Onion powder	0.15
Pepper, black, ground	0.05

The gravy shall be prepared as follows:

a. A slurry shall be made with the starch and part of the water. The balance of the water, the lecithin, and the vegetable shortening shall be heated to a boil in a steam jacketed kettle with continuous and vigorous mixing to attain maximum emulsification of the fats into the water. The emulsion may be cooled to 180°F or less and the mushrooms and the remaining ingredients added and uniformly mixed in to dissolve the solubles. The starch slurry shall be added and the final mixture heated to 180° to 190°F and held for approximately 5 minutes at this temperature.

b. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating.

c. The gravy shall be filled within 4 hours after cooking. The temperature of the prepared gravy shall not be allowed to fall below 150°F at any time prior to filling into the cans.

d. Alternatively, canned mushrooms may be spread separately over the beef slices before adding the gravy in the filling process rather than adding in combination with the gravy, as long as the end-item requirements are met.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product so as to conform to the finished product requirements and to the following requirements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

a. Not less than 40 intact slices (each approximately 1-3/4 by 4-1/2 inch) of roast beef shall be shingled in two rows in the tray pack can, and the gravy shall be added to meet the net weight requirement of 108 ounces.

b. The temperature of the beef at time of filling shall be 28°F to 40°F.

c. The temperature of the gravy at time of filling shall be not less than 150°F.

d. Each can shall be hermetically sealed (under a vacuum of not less than 10 inches of mercury) immediately after filling.

e. The filled and sealed tray pack cans shall be in the retort process within 1 hour after filling.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting at 250°F or below until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material, such as, but not limited to dirt, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

f. No individual can shall contain less than 40 approximately 1-3/4 by 4-1/2 inch intact beef slices.

g. No individual can shall contain less than 2.0 ounces drained weight of mushrooms.

h. There shall be no pockets of fat in beef slices greater than 1/2 inch in any dimension extending through the slice.

i. No individual can shall contain less than 106 ounces of product.

j. The average net weight shall be not less than 108 ounces.

k. Forty slices of roast beef shall weigh not less than 56 ounces.

l. The gravy consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

m. Beef slices shall not be dry, tough, rubbery, or mushy.

n. Beef slices shall be shingled in the tray can.

o. The average fat content of the finished product shall be not greater than 15.0 percent, and no individual tray pack shall have a fat content which is greater than 17.0 percent.

p. No individual tray pack shall have a salt content greater than 1.2 percent.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or other applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.4.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts. Any nonconforming beef shall be rejected.

4.4.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size expressed in terms of pounds shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (No. sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	13	1	2	3	4
501 to 1200	20	2	3	5	6
1201 to 3200	32	3	4	7	8
3201 to 10000	50	5	6	10	11
10001 to 35000	80	7	8	14	15
Reinspection lots of 10001 to 35000	125	10	11	21	22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE II. Boned and trimmed beef defects

Category		Defect 1/ 2/
Major	Minor	
101		Presence of lymph gland measuring 1.0 inch or more in any dimension.
	201	Presence of lymph gland measuring 0.5 inch or more but less than 1.0 inch in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension.
	202	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.
	203	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	204	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin or the gracilis membrane) measuring 2.0 square inches or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	205	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin or gracilis membrane) measuring 0.5 square inches or more but less than 2.0 square inches.
	206	Presence of heavy connective tissue on lower edge of short plate or flank.
	207	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	208	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3 square inches or more.
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring less than 1.0 inch in longest dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other other).
	210	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	211	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	212	Presence of dehydrated surface measuring 1.0 square inch or more.
	213	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE II. Boned and trimmed beef defects (cont'd)

Category		Defect 1/ 2/
Major	Minor	
	214	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension.
	215	Presence of cod, udder, kidney, or pelvic fat.

1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g.; glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the lot.

4.4.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

to the nearest 1 ounce. If the average net weight is less than 108 ounces or if the net weight of any individual can is less than 106 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated at least 30 minutes in 160° to 180°F water, opened, and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8	-	0	2
	8	16	1	2
3201-35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Less than 40 intact beef slices (each approximately 1-3/4 by 4-1/2 inch) per tray can.
102	Drained weight of mushrooms in a can is less than 2.0 ounces. 3/
103	Presence of fat pocket measuring greater than 1/2 inch in any dimension and extending through the slice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

TABLE IV. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
104	Gravy consistency is not smooth, is lumpy, or is excessively thick or thin.
105	Beef slices dry, tough, rubbery, or mushy.
106	Slices not arranged in a shingled manner.
107	Forty slices of roast beef weigh less than 56 ounces. 3/

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquify rendered fat and remove the gravy without breaking the roast beef slices. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the roast beef slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. After total drained weight has been determined, remove the mushrooms and reweigh. The drained weights shall be reported to the nearest 0.1 ounce.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.6.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- three for laboratory analysis.
- three for submission to the contractor.
- three for retention by an AMS agent as a reserve sample.

The three tray packs for laboratory analyses shall individually be tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows: The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The lot shall be rejected if:

a. The average fat content of the three tray packs is greater than 15.0 percent.
b. The fat content of any individual tray pack is greater than 17.0 percent.
c. The salt content of any individual tray pack is greater than 1.2 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: (1) National stock number, item description, contract number, or date of pack markings missing, incorrect or illegible; (2) Reinforcement with nonmetallic strapping or tape is not used; Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (4) Length or width of pads less than box dimensions by more than 1/8 inch; (5) Arrangement or number of cans not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 108 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (See 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 35 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of tray can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately twenty (20) servings of 2 slices each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Suggested cookhouse method for meat log cooking. During production testing of the product, the following cooking method was found to yield an acceptable product:

a. Cook 1 hour with wet bulb temperature of 130°F and dry bulb temperature of 150°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

b. Continue cooking for 2 hours with wet bulb temperature of 150°F and dry bulb temperature of 170°F.

c. Continue cooking with dry bulb temperature of 180°F (without steam or moisture) until the meat logs reach an internal temperature of 155°F.

Preparing activity

Army - GL

Project No. 8940-A502

U.S. GOVERNMENT PRINTING OFFICE: 1986 - 605-036/41090

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

US ARMY NATICK RESEARCH
and DEVELOPMENT CENTER
ATTN: STRNC-ES
Natick, MA 01760-5014

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research
and Development Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
EE-B-002181A(Army-GL)	Beef, Roast, with Mushroom Gravy, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:44Z
Check the source to verify that this is the current version before use.

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 20 Nov 84